1.1.25

Vishwambhar - EE25BTECH11025

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Question

The state-space representation of a separately excited DC servo motor dynamics is given as:

$$\frac{d}{dt}\begin{pmatrix}\omega \\ i_a\end{pmatrix} = \begin{pmatrix}-1 & 1 \\ -1 & -10\end{pmatrix}\begin{pmatrix}\omega \\ i_a\end{pmatrix} + \begin{pmatrix}0 \\ 10\end{pmatrix} u \tag{1}$$

where ω is the speed, i_a is the armature current, and u is the armature voltage.
The transfer function $\frac{\omega(s)}{u(s)}$

1. $\frac{10}{s^2+11s+11}$
2. $\frac{1}{s^2+11s+11}$
3. $\frac{10s+10}{s^2+11s+11}$
4. $\frac{1}{s^2+s+1}$

Given

$$\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}u \tag{2}$$

where,

$$\mathbf{x} = \begin{pmatrix} \omega \\ i_a \end{pmatrix} \tag{3}$$

$$A = \begin{pmatrix} -1 & 1 \\ -1 & -10 \end{pmatrix} \tag{4}$$

$$\mathbf{b} = \begin{pmatrix} 0 \\ 10 \end{pmatrix} \tag{5}$$

Taking Laplace Transform

$$sX(s) = AX(s) + \mathbf{b}U(s) \tag{6}$$
$$(sI - A)X(S) = \mathbf{b}U(s) \tag{7}$$
$$X(s) = (sI - A)^{-1} + \mathbf{b}U(s) \tag{8}$$

we only want ω, so:

$$Y(s) = \mathbf{c}\top X(s) \tag{9}$$
$$Y(s) = \mathbf{c}\top (sI - A)^{-1} bU(s) \tag{10}$$

where,

$$c = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{11}$$

Taking Laplace Transform

$$sX(s) = AX(s) + \mathbf{b}U(s) \tag{12}$$
$$(sI - A)X(S) = \mathbf{b}U(s) \tag{13}$$
$$X(s) = (sI - A)^{-1} + \mathbf{b}U(s) \tag{14}$$

we only want ω, so:

$$Y(s) = \mathbf{c}\top X(s) \tag{15}$$
$$Y(s) = \mathbf{c}\top (sI - A)^{-1} bU(s) \tag{16}$$

where,

$$c = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{17}$$

Computing

Final formula,

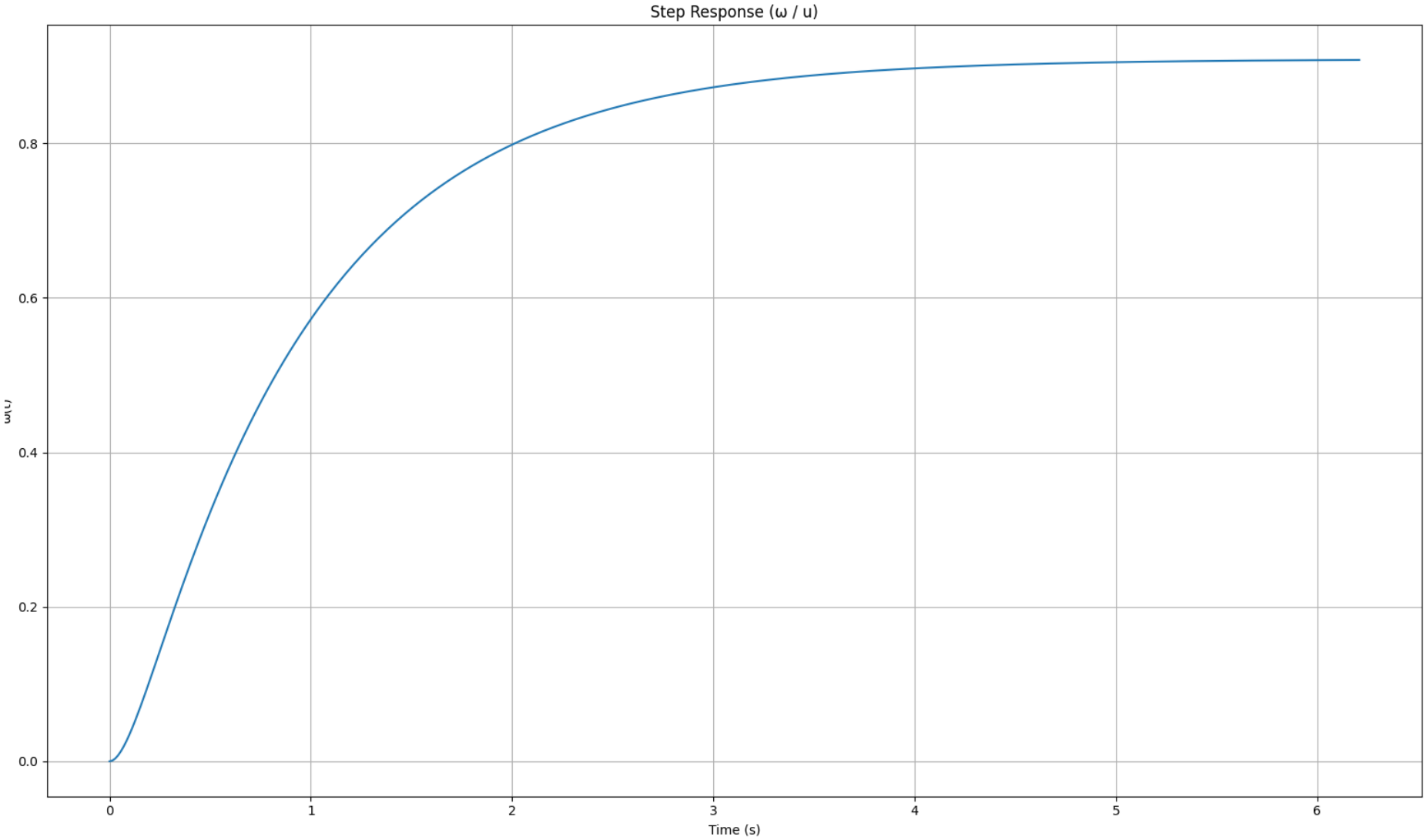
$$G(s) = \frac{Y(s)}{U(s)} = \mathbf{c}\top (sI - A)^{-1} b \tag{18}$$

Multiplying Everything,

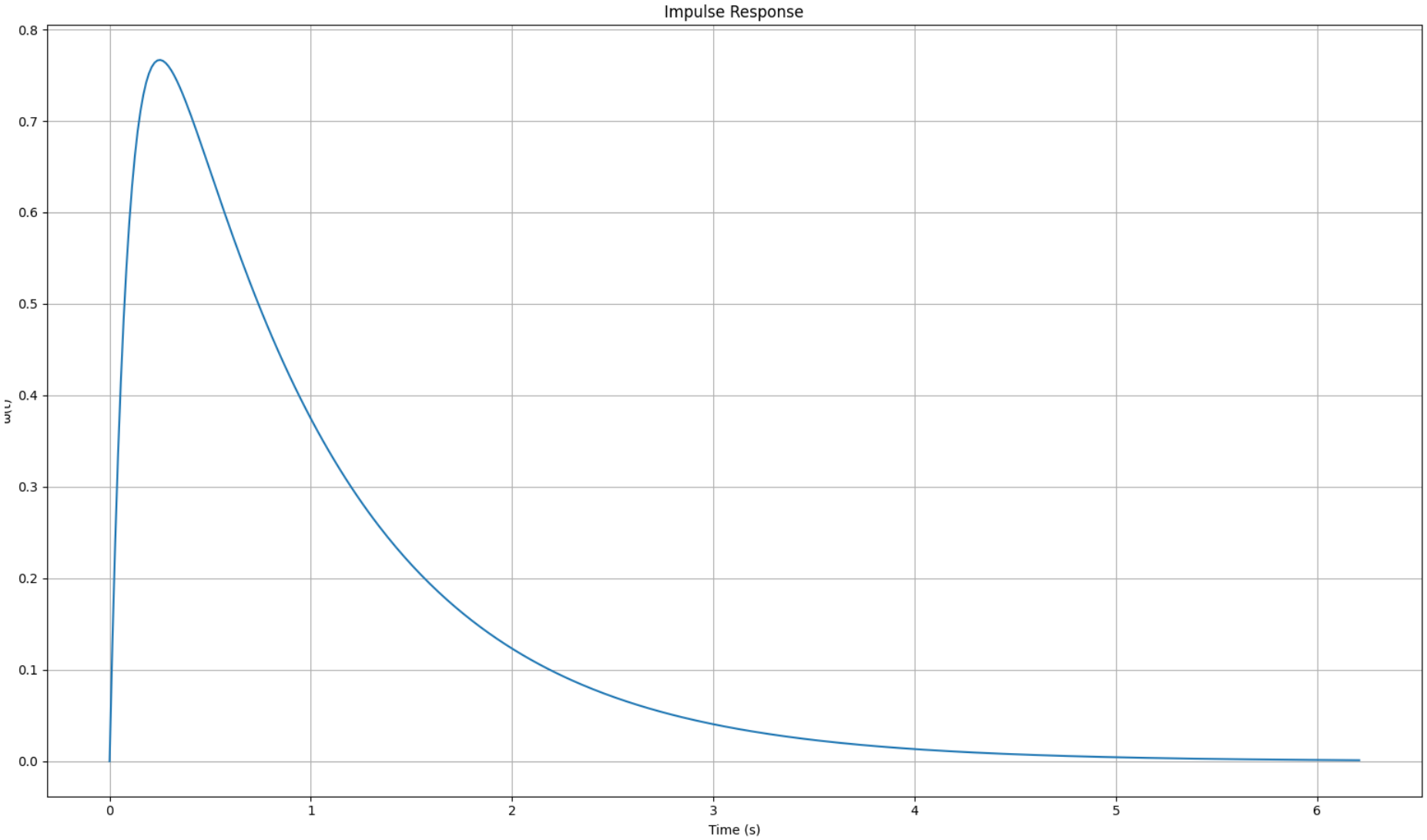
$$(sI - A)^{-1} b = \frac{1}{s^2 + 11s + 11} \begin{pmatrix} s + 10 & 1 \\ -1 & s + 1 \end{pmatrix} \begin{pmatrix} 0 \\ 10 \end{pmatrix} \tag{19}$$

$$= \frac{1}{s^2 + 11s + 11} \begin{pmatrix} 10 \\ 10(s + 1) \end{pmatrix} G(s) = \frac{10}{s^2 + 11s + 11} \tag{20}$$

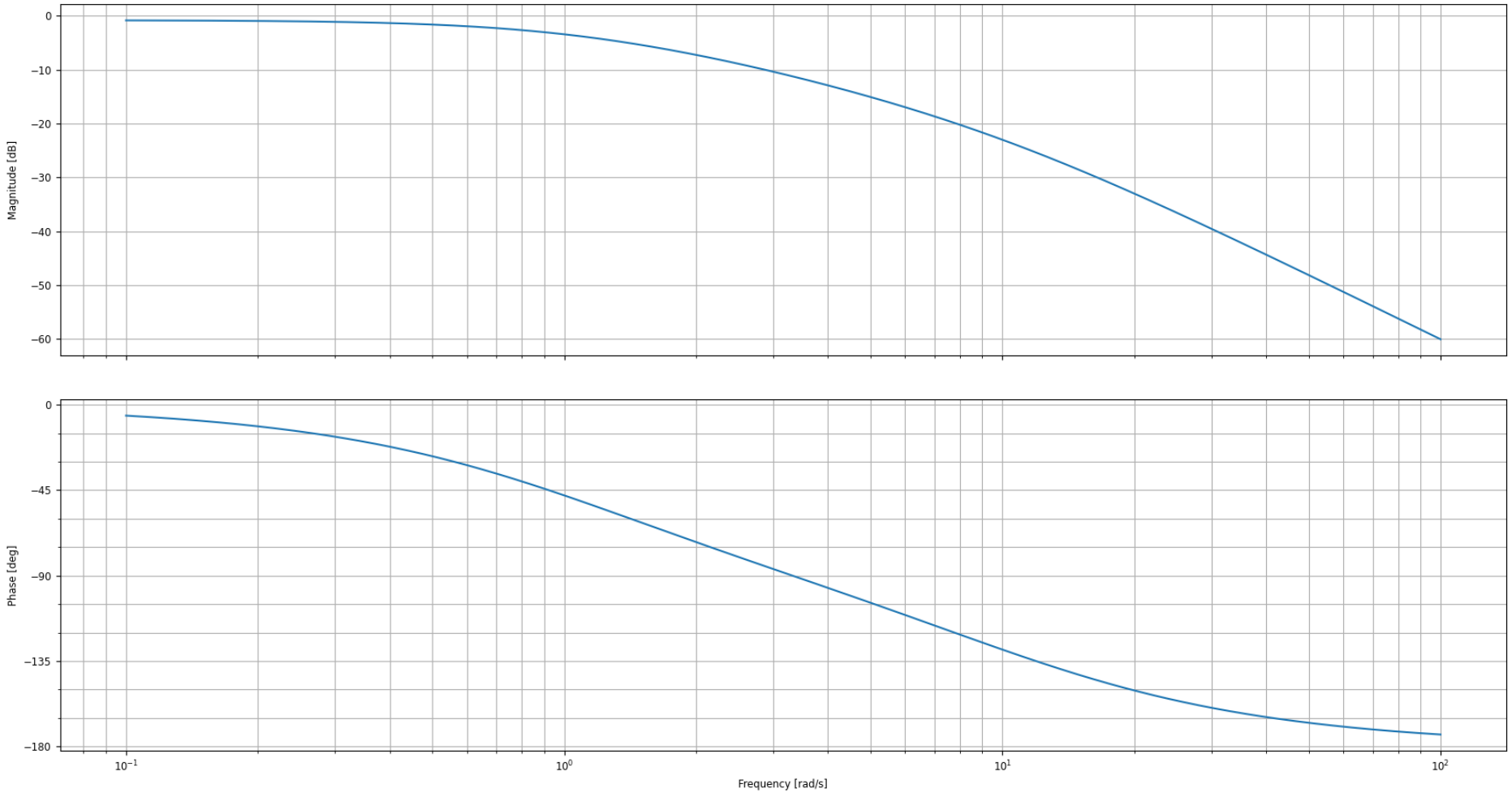
Plot-Step Response



Plot-Impulse Response



Plot-Sine Input(Bode)



Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import control as ctrl

# Define matrices
A = np.array([[-1, 1],
              [-1, -10]])
B = np.array([[0],
              [10]])
C = np.array([[1, 0]])
D = np.array([[0]])

sys_ss = ctrl.StateSpace(A, B, C, D)
sys_tf = ctrl.ss2tf(sys_ss)

print(Transfer Function:)
print(sys_tf)
```

Python Code

```
# STEP RESPONSE
t, y = ctrl.step_response(sys_tf)
plt.figure()
plt.plot(t, y)
plt.title(Step Response ( / u))
plt.xlabel(Time (s))
plt.ylabel((t))
plt.grid()
```

Python Code

```
# IMPULSE RESPONSE
t, y = ctrl.impulse_response(sys_tf)
plt.figure()
plt.plot(t, y)
plt.title(Impulse Response)
plt.xlabel(Time (s))
plt.ylabel((t))
plt.grid()

# BODE PLOT
plt.figure()
ctrl.bode(sys_tf, dB=True)

plt.show()
```